BITCOIN WEB-OF-TRUST: SNA APPLIED ASSESSMENT TASK • DEPARTMENT OF MACHINE LEARNING

Team Presentation & Viva Study Guide (3-Speaker Plan)

The 60-Second Elevator Pitch: In OTC cryptocurrency markets, anonymous traders face extreme fraud risk because trades are irreversible and have no escrow. We formulate trust recommendation as a topological link prediction task on the Bitcoin Web-of-Trust. Using four fundamental heuristics (Common Neighbors, Jaccard, Adamic-Adar, Preferential Attachment) and a normalized rank ensemble, our models predict future trading partners with 0.9505 ROC-AUC and 100% Precision@20, evaluated across 9,207 real traders (~60k ratings) and a 25,000-node enterprise synthetic network. 100% of recommended ties satisfy Heider's Structural Balance by completing (+, +, +) triads.

1. Who Presents What: 3-Speaker Slide Division across 14 Slides

SPEAKER 1: Slides 1 – 5 (Problem Definition, Data Engineering, Scale-Free Topology & 80/20 Split)

• *Slide 1:* Introduce the team, project title, and structure.
• *Slide 2:* Explain OTC counterparty risk, anonymous trading, ratings from -10 to +10, and why link prediction is the ideal formulation.
• *Slide 3:* Explain that we combined Stanford SNAP Bitcoin Alpha (3,783 users) and Bitcoin OTC (5,887 users) into 9,207 real traders and ~60k ratings.
• *Slide 4:* Present the log-log scale-free degree distribution fit (P(k) ~ k^(-2.1)) and explain why broker hubs drive Preferential Attachment accuracy.
• *Slide 5:* Present the 25k-node enterprise synthetic scaling (Holme-Kim model), 80/20 train-test split, Minimum Spanning Tree protection, and balanced 50/50 negative sampling.

SPEAKER 2: Slides 6 – 9 (Link Prediction Algorithms & Mathematical Formulations)

• *Slide 6:* Introduce the philosophy of topological heuristics and triadic closure (a friend of my friend is my friend).
• *Slide 7:* Explain Common Neighbors ($|\Gamma(u) \cap \Gamma(v)|$; 'mutual friends') and Jaccard (|Common| / |Union|; 'neighborhood overlap %').
• *Slide 8:* Explain Adamic-Adar (log degree penalty; 'exclusive introductions'), Resource Allocation (linear degree penalty), and Preferential Attachment (deg_u × deg_v; 'market maker liquidity hub' analogy; 0.9505 AUC).
• *Slide 9:* Present the Normalized Rank Ensemble Model combining 30% AA + 25% RA + 25% Jaccard + 10% CN + 10% PA.

SPEAKER 3: Slides 10 – 14 (Benchmark Results, Top-20 Demo, Community Roles & Viva)

• *Slide 10:* Present the test scoreboard: Preferential Attachment (0.9505 AUC), Ensemble (0.9387 AUC), and 100% Precision@20 on local heuristics.
• *Slide 11:* Present the Enterprise Scaling chart: 0.7539 AUC and 100% Precision@10 and @20 on 25k nodes.
• *Slide 12:* Walk through the live Top-20 recommendations for Trader #1. Explain Community Roles (Intra-Cluster vs Inter-Cluster Bridge) and prove 100% (+, +, +) triad closure under Heider's Structural Balance Theory.
• *Slide 13:* Showcase the interactive web UI and complete checklist of the 5 official course deliverables.
• *Slide 14:* Deliver the viva defense summary and conclude the presentation.

2. Key Numbers to Memorize for Viva

- **Real Data:** 9,207 traders, 31,050 verified ties, ~60,000 ratings (Alpha + OTC combined).
- **Enterprise Synthetic Data:** 25,000 nodes, 135,857 ties (Holme-Kim power-law clustering model).
- **Train-Test Split:** 80% train / 20% test with Minimum Spanning Tree protection.
- **Evaluation Size:** 3,647 positive test ties + 3,647 negative non-edges = 7,294 test instances.
- **Top ROC-AUC:** 0.9505 (Preferential Attachment) and 0.9387 (Ensemble Model).
- **Top-20 Precision:** 100.0% across PA, AA, and Common Neighbors.
- **Structural Balance:** 100% of recommended links form balanced (+, +, +) triads.

3. Top 5 Viva Questions & Word-for-Word Answers

Q1: Why does Preferential Attachment have such a dominant ROC-AUC (0.9505)?
Answer: Financial OTC networks are strongly scale-free (gamma ≈ 2.1). In cryptocurrency trading, participants route trades through established, reputable brokers who act as liquidity hubs. Because PA multiplies node degrees, it naturally captures this hub attraction, providing near-perfect global ranking.

Q2: If PA has the highest AUC, why bother with Adamic-Adar or Common Neighbors?
Answer: Preferential Attachment has zero community specificity—it just recommends the biggest hub to everyone. Adamic-Adar specifically looks at who your mutual connections are and penalizes hubs. It identifies exclusive, tight-knit community partners, achieving 100% Precision@10.

Q3: Why was Spanning Tree protection mandatory in the 80/20 train-test split?
Answer: In sparse scale-free networks, randomly deleting 20% of edges frequently disconnects peripheral nodes into isolated components. Isolated nodes have zero common neighbors, creating artificial false negatives. Retaining an MST in G_train ensures the graph remains fully connected.

Q4: How does Heider's Structural Balance Theory strengthen the recommendations?
Answer: In signed trust networks, cognitive tension occurs in unbalanced triads like (+, +, -) ('my friend trusts my enemy'). We verified that 100% of our top recommended links close (+, +, +) triads with mutual partners, ensuring newly introduced connections reinforce community stability.

Q5: What are the primary deliverables of your project?
Answer: 1. Source code & executed notebook (.ipynb); 2. Dataset package with SHA-256 manifest; 3. Visualizations (ROC curves, Precision@K bar charts, Cytoscape GraphML, HTML demo); 4. 8-page academic report (PDF); 5. 13-slide 3-speaker presentation deck (.pptx and .pdf); plus a local interactive web UI.